

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Sixth Semester B.E. (E&E) (Credit System) Degree Examinations**

May - June 2023

20EE603 - HIGH VOLTAGE ENGINEERING

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I

	Marks	BT*	CO*	PO*
1. a) What are the advantages of transmitting electrical power at high voltage levels.	6	L*2	1	1,2,6
b) Enumerate the properties of high voltage insulating materials.	6	L2	1	1,2,6
c) Explain the Townsends first ionisation coefficient. Illustrate the current growth equation and derive criteria for break down in gases.	8	L2	1	1,2,6
2. a) Explain the thermal break down in liquid dielectrics.	6	L2	2	1,2,6
b) Explain Paschen's law with the help of graph.	6	L2	2	1,2,6
c) Explain Townsends second ionization coefficient and derive Current growth equation.	8	L2	2	1,2,6
3. a) Explain the cavity break in liquid dielectrics.	6	L2	1	1,2,6
b) Explain the streamer theory of break down in gases.	6	L2	1	1,2,6
c) The following Observations were made in an experiment for determination of dielectric strength of transformer oil. Determine the power law equation.				
Gap Spacing	4	6	8	10
Break downs Voltage (KV)	88	135	165	212
	8	L3	2	1,2,6

Unit – II

4. a) With a neat circuit, explain the series Resonant circuit and enumerate the advantages of it.	7	L2	3	1,2,6
b) Explain the principle of voltage doubler circuit used for HVDC generation.	6	L2	3	1,2,6
c) A 100 KVA, 250V/200KV feed transformer has resistance and reactance of 1% & 5% respectively. This transformer is used to test a cable at 400 kV at 50 Hz. The cable takes a charging current of 0.5A at 400KV. Determine the series inductance required. Assume 1% resistance of the inductor. Also determine input voltage to the transformer. Neglect the dielectric loss of cable.	7	L3	3	1,2,6
5. a) With a neat circuit, explain generation of high voltage ac using cascaded transformers.	10	L2	3	1,2,6
b) A 10 stage cockraft Walton circuit has all capacitors of 0.06 μ F. The secondary voltages of the supply transformer is 100kV at a frequency of 150Hz. If the load current is 1mA, determine i) voltage regulation ii) The ripple iii) Optimum number of stages for maximum output voltage iv) Maximum output voltage.	10	L3	3	1,2,6

P.T.O.

6. a) Explain with neat circuit triggering and synchronization of the impulse generator with CRO. 10 L2 4 1,2,6
- b) A 12 stage impulse generator has capacitors each rated at $0.3 \mu\text{F}$, 150 kV. The capacitance of the test specimen is 400 Pico Farad. Determine the Wave front and Wave tail resistance to produce at $1.2 / 50 \mu\text{sec}$ impulse wave. Also determine the maximum output voltage if the charging voltage is 125 kV. 10 L3 4 1,2,6

Unit – III

7. a) Explain with neat schematic diagram, the principle of operation of Electrostatic voltmeter. Discuss its advantages and disadvantages. 10 L2 5 1,2,3
- b) A Rogowski Coil is required to measure impulse current of 8 kA having rate of change of current of 10^{10} A/sec . The voltmeter is connected across the integrating circuit which heads 8 Volts for full scale deflection. The input to the integrating circuit is from the Rogowski Coil. Determine the mutual inductance of the coil R&C of the integrating circuit. 10 L3 5 1,2,3
8. a) Draw the Chubb-Fortescue circuit of measurement of peak voltage of ac voltages. Discuss its advantages over other methods. 12 L2 5 1,2,6,7
- b) An absolute Electrostatic voltmeter has a movable circuit plate 8 cm in diameter. If the distance between the plates during a measurement is 4mm, determine the potential difference when the force of attraction is 0.2 gm wt. 8 L3 5 1,2,6,7

BT* Bloom's Taxonomy, L* Level, CO* Course Outcome, PO* Program Outcome.
